

Lavender Elle Hanson

Postdoctoral Researcher
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Research positions

Modeling PAH in Titan's atmosphere 2025–
University of Amsterdam
Advisor: Alessandra Candian

Faculty Research Associate 2020
Laboratory spectroscopy of Titan cloud ices
NASA Goddard Space Flight Center/University of Maryland
advised by Carrie M. Anderson

Faculty Research Associate 2019
Martian cloud and atmospheric dynamics using Mars Reconnaissance Orbiter imagery
NASA Goddard Space Flight Center/University of Maryland
advised by Scott Guzewich

Education

Ph.D., Earth and Planetary Science 2020–2025
Stratospheric ices in Titan's south polar fall, modeling and imagery analysis
Johns Hopkins Univ, Baltimore, MD, USA
Advisor: Darryn D. Waugh

M.S., Atmospheric Science & Meteorology 2013–2018
Theoretical and experimental microphysics of water ice clouds
Penn State Univ, University Park, PA, USA
Advisor: Jerry Y. Harrington

B.A., Physics & Chemistry 2009–2013
Luther College, Decorah, IA, USA

Awards

- Johns Hopkins Krieger School of Arts & Sciences Dissertation Prize Fellowship 2025

Mentorship & supervision

Bachelor's thesis project supervisor Spring 2026

ASPIRE (Astrophysics Summer Program for International Research Experience) Summer 2026

- Mentoring a MSc student completing a short research project

HACK-4-SAGES Astrobiology Hackathon, Amsterdam, NL March 2026

- Supervised and advised a team of undergraduate students

Teaching

- Guided Tour: The Planets, TA 2023 & 2024

- Principles of Atmospheric Measurement, co-instructor Spring 2017
- Radiation and Climate, TA Fall 2018
- Atmospheric Thermodynamics, TA Fall 2014

Teaching Training

- JHU teaching academy workshops 2020–2025
- Johns Hopkins Teaching Institute, May 29-31 Spring 2024
- “Grading and Anti-grading” (workshop series) Spring 2024
- “Teaching Discomfort: Facilitating Challenging Discussions in the Classroom” (workshop series) Fall 2023

Service

- UvA/API representative to NOVA (Dutch Schools for Astronomy) 2025–
- Postdoc council representative, API 2025–2026
- Newsletter Editor and Contributor, EPS 2024–2025
- Social Committee organizer, EPS 2020–2024
- Johns Hopkins Trans Awareness Task Force 2023–2024

Professional Society Membership

- Europlanet Society 2025–present
- American Astronomical Society, Division of Planetary Science 2023–present
- American Geophysical Union 2018–present

Funding

- NASA FINESST: *Mixed-species clouds in Titan's polar stratosphere* (as future investigator, PI: Darryn Waugh). 2021–2024

Publications

- Lavender E Hanson, Robert French, Darryn Waugh, Erika Barth, and Carrie M. Anderson, 2025: The Descent of Titan’s South Polar Cloud, *Geophys Res Lett*, 52, e2024GL113415. doi:[10.1029/e2024GL113415](https://doi.org/10.1029/e2024GL113415)
- Lavender E Hanson, Darryn Waugh, Erika Barth, and Carrie M. Anderson, 2023: Investigation of Titan's south polar HCN cloud during southern fall using microphysical modeling, *Planet Sci J*, 4, 237. doi:[10.3847/PSJ/ad0837](https://doi.org/10.3847/PSJ/ad0837)
- Gwenore F Pokrifka, AM Moyle, Lavender E Hanson, and Jerry Y Harrington, 2020: Estimating Surface Attachment Kinetic and Growth Transition Influences on Vapor-Grown Ice Crystals, *J Atmos Sci*, 77, 2393. doi:[10.1175/jas-d-19-0303.1](https://doi.org/10.1175/jas-d-19-0303.1)
- Jerry Y Harrington, Alfred Moyle, Lavender E Hanson, Hugh Morrison, 2019: On Calculating Deposition Coefficients and Aspect-Ratio Evolution in Approximate Models of Ice Crystal Vapor Growth, *J Atmos Sci*, 76, 1609. doi:[10.1175/jas-d-18-0319.1](https://doi.org/10.1175/jas-d-18-0319.1)

- Alexander Harrison, Alfred M Moyle, Hanson, Jerry Y Harrington, 2016: Levitation diffusion chamber measurements of the mass growth of small ice crystals from vapor, *J Atmos Sci*, 73, 2743-2758. doi:[10.1175/JAS-D-15-0234.1](https://doi.org/10.1175/JAS-D-15-0234.1)
- EM Levin, R Hanus, Hanson, WE Straszheim, K Schmidt-Rohr, 2013: Thermoelectric properties of $\text{Ag}_2\text{Sb}_2\text{Ge}_{46-x}\text{Dy}_x\text{Te}_{50}$ alloys with high power factor, *Physica Status Solidi A*, 210, 2628-2637. doi:[10.1002/pssa.201330217](https://doi.org/10.1002/pssa.201330217)

Invited talks

- “Clouds and chemistry on Titan,” Nederlandse Onderzoekschool Voor Astronomie (NOVA) Network II Meeting, Amsterdam, Netherlands, 28 November.
- “Chemistry and clouds on Saturn’s moon Titan,” Amsterdam Center for Origins Of Life (AMCOOL) Symposium, Amsterdam, Netherlands, 10 November.

Conference presentations

- Lavender E Hanson, Alessandra Candian, Panayotis Lavvas, Veronique Vuitton, 2026: Producing naphthalene in Titan’s atmosphere (poster). *Exploring the Aromatic Universe in the JWST Era*, London, Ontario, 2.18.
- Lavender E Hanson, D W Waugh, C Anderson, E Barth, and R French, 2025: Evolution of Titan’s Fall and Winter Polar Hood Cloud (talk). *EPSC-DPS Joint Meeting 2025*, Helsinki, Finland, EPSC-DPS2025-221, doi: [10.5194/epsc-dps2025-221](https://doi.org/10.5194/epsc-dps2025-221)
- Lavender E Hanson, E L Barth, C M Anderson, and D W Waugh. 2025: Benzene and HCN Ice Clouds in Titan’s Polar Stratosphere (talk). *LPSC 2025*, The Woodlands, TX, #1521.
- Lavender E Hanson, Darryn Waugh, Erika Barth, and Carrie Anderson. 2024: Observations and Modeling of Titan’s Descending South Polar Cloud (poster). *AGU 2024*, Washington, DC, P43D-3038.
- Lavender E Hanson, Darryn Waugh, Carrie Anderson, and Erika Barth. 2024: The Slow Descent of Titan’s South Polar Cloud (talk). *AAS/DPS 2024*, 208.02, Boise, ID.
- Lavender E Hanson, Darryn Waugh, Erika Barth, and Carrie M. Anderson. 2023: Investigating the evolution of Titan's high altitude south polar HCN cloud (talk). *AAS/DPS 2023*, 208.04, San Antonio, TX.
- Lavender E Hanson, Darryn Waugh, Erika Barth, and Carrie M. Anderson. 2023: Modeling the fall high altitude south polar HCN cloud (talk). *Titan Through Time 6*, Paris.
- Lavender E Hanson, Scott Guzewich, 2019: Orographic clouds in the Mars Arcadia province (poster). *AGU Fall Meeting 2019*, P41B-3405.
- Lavender E Hanson, Scott Guzewich, 2019: Using Machine Learning to Identify Clouds in Mars Daily Global Maps (poster), *Ninth International Conference on Mars*, Pasadena, CA.
- Hanson, Alfred Moyle, Jerry Harrington, 2016: Measurements of vapor growth and sublimation of individually levitated ice particles below -30°C (talk), *17th International Conference on Clouds & Precipitation*, Manchester, UK, S1.14.

Updated: August 17, 2026